

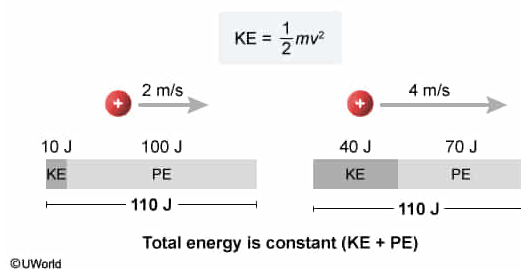
A charged particle is accelerated in a uniform electric field. When its velocity is 2 m/s, its electric potential energy is 100 J and its kinetic energy is 10 J. What is the particle's potential energy when its velocity reaches 4 m/s?

- ☐ A. 25 J [12%]
✓ ☒ B. 70 J [58%]
☐ C. 90 J [19%]
☐ D. 135 J [8%]

Correct

58% answered correctly

Explanation:



Electric charges generate electric fields, regions of space that exert an **electrostatic force** on charges. The electrostatic force is conservative and the **total energy** E of a charged particle in an electric field is **conserved**. In other words, the sum of the charge's **electric potential energy** (PE) and **kinetic energy** (KE) is constant:

$$E_{\text{initial}} = E_{\text{final}}$$

$$PE_{\text{initial}} + KE_{\text{initial}} = PE_{\text{final}} + KE_{\text{final}}$$

The initial E of the particle in the electric field is the sum of the given PE and KE :

$$E_{\text{initial}} = PE_{\text{initial}} + KE_{\text{initial}}$$

$$E_{\text{initial}} = 100 \text{ J} + 10 \text{ J} = 110 \text{ J}$$

Furthermore, KE is proportional to mass m and the **square of velocity** v :

$$KE = \frac{1}{2}mv^2$$

As the particle accelerates, its velocity increases by a factor of 2, from 2 to 4 m/s. Therefore, its final KE increases by a factor of 4:

$$KE_{\text{final}} = \frac{1}{2}m(2v)^2 = 2^2\left(\frac{1}{2}mv^2\right) = 4\left(\frac{1}{2}mv^2\right)$$

$$KE_{\text{final}} = 4(KE_{\text{initial}}) = 4(10 \text{ J}) = 40 \text{ J}$$

The final PE of the particle at 4 m/s is the difference between the overall total energy and its final KE :

$$PE_{\text{final}} = E_{\text{final}} - KE_{\text{final}} = E_{\text{initial}} - KE_{\text{final}}$$

$$PE_{\text{final}} = 110 \text{ J} - 40 \text{ J} = 70 \text{ J}$$

(Choice A) The *sum*, not the product, of *PE* and *KE* for a conservative force is constant. If 1,000 J is incorrectly used as *KE*, the calculated *PE* would be $1,000/40 = 25 \text{ J}$.

(Choice C) The *KE* of the particle should increase by a factor of 4, not 2, when its velocity increases by a factor of 2. If 20 J is incorrectly used as its *KE*, the calculated *PE* would be 90 J.

(Choice D) The *PE* of the particle cannot be greater than its sum of *PE* and *KE* because *KE* cannot be negative.

Educational objective:

Electrostatic force is conservative; the sum of potential energy and kinetic energy is constant.

KE is the product of mass m and the square of velocity v : $KE = \frac{1}{2}mv^2$. Therefore, the potential energy of a particle is the difference between its total energy and its *KE*.

Time spent: 0 sec

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